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Trends and challenges in the operation of pumped-storage hydropower plants

Juan I. Pérez-Díaz [a,*], M. Chazarra [a], J. García-González [b], G. Cavazzini [c], A. Stoppato [c]

[a] Department of Hydraulic and Energy Engineering, Technical University of Madrid (UPM), c/Profesor Aranguren s/n, 28040 Madrid, Spain
[b] Institute for Research in Technology, Comillas Pontifical University, c/Alberto Aguilera 23, 28015 Madrid, Spain
[c] Department of Industrial Engineering, University of Padova, Via Venezia 1, Via Gradenigo, 6/a – 35131 Padova, Italy

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ABSTRACT

The increasing penetration of renewable energy sources (RESs) in the power system has highlighted the benefits of being able to store energy in a more efficient manner, and the need of holding additional operating reserves to manage the system under more demanding conditions due to the inherent uncertainty and variability of wind and solar power. Pumped hydroelectric energy storage (PHES) is by far the most established technology for energy storage at a large-scale. PHES units have also participated in the active power-frequency control for years, and last technical developments in PHES have been oriented to improve their capability of providing regulation reserves by means of variable speed design or by hydraulic short-circuit configurations. This fact, together with the impact of RES on spot-market energy and ancillary services prices, is changing the operation and market conditions faced by PHES plants. The aim of this paper is to review the current trends in the PHES operation, to discuss why current practices should be re-examined, and to present the main challenges faced by PHES operators who will need to adapt their scheduling and bidding models to optimize jointly the operation in the energy and in the ancillary services markets.

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Contents

1. Introduction . . . 767
2. Trends of pumped-storage hydropower plants operation . . . 768
   2.1. Multi-market operation . . . 768
   2.2. Modeling and management of market uncertainties . . . 771
   2.3. Wind-hydro coordination . . . 774
   2.4. Solutions for providing power regulation in consumption mode (variable speed and short-circuit operation) . . . 776
   2.5. Summary of observed trends . . . 776
3. Challenges . . . 778
   3.1. Considering the uncertainty of the real time use of regulation reserves . . . 778
   3.2. Revising the methods currently used to determine the long-term guidelines (target stored volumes and marginal water value functions) . . . 778
   3.3. Defining the optimal strategy for PSHPs with variable speed and hydraulic short-circuit operation . . . 779
4. Conclusions . . . 781
Acknowledgments . . . 781
Appendix Acronyms used throughout the paper (in alphabetical order): . . . 781
References . . . 782

1. Introduction

Among the available technologies to store energy at a large-scale level, pumped hydroelectric energy storage (PHES) is the most widely adopted one. The big amount of potential energy that

* Corresponding author. Tel.: +34 913366705.
E-mail address: ji.perez@upm.es (J.I. Pérez-Díaz).

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can be stored in hydro reservoirs, the energy conversion efficiency of the whole cycle, the cost per power unit, and the flexibility provided by these plants to the Transmission System Operator (TSO) in the short-term operation makes PHES the most attractive option for large-scale energy storage. As a consequence, pumped-storage hydropower plants (PSHPs) have been widely installed and operated since the 1890s, reaching an approximate worldwide installed capacity of 130 GW [1].

The traditional operation of PSHPs is mainly focused on satisfying the load by means of the so called hydro-thermal coordination. Thus, the water is pumped during off-peak hours when the demand is low, and it is released afterwards during peak-hours with an overall round-trip efficiency in the range of 70–80%. This cycle can be daily, weekly, or even seasonal when the plants are located in not-isolated hydro systems. In power systems organized as electricity markets, the operation of the generators depends on the results of the market clearing, and only an adequate pricing mechanism can lead to an optimal use of the generation resources.

Due to the well-known environmental concerns, and thanks to a number of different renewable energy sources (RESs) support policies [2,3], wind and solar power have increased notably their market share in many power systems during the last decade. Amongst all RESs, wind seems to be at present the one with the largest economically feasible potential [4], but also with the highest variability [5,6], and the most difficult to predict [7]. Therefore, the presence of high levels of wind power in the generation mix is challenging the operation of power systems. On the one hand, wind power inherent variability and uncertainty require more advanced scheduling models, having a direct impact on the amount and type of the required operation reserves [8]. On the other hand, both wind and solar power generators cannot decide to defer their generation as they are subject to the instantaneous meteorological conditions. Thus, RES curtailments can occur in case of having an excess of supply (particularly during off-peak hours) when conventional generators cannot reduce their output due to technical constraints. Increasing the storage capability, both at a local and at a system level, is a natural way to mitigate this problem [9,10].

Apart from the technical aspects, the increasing penetration of RES in power systems is also affecting the resulting spot-market prices. It has been reported that besides the impact on the average energy prices [11], RES can also affect the shape of the hourly price profiles [12]. As the profitability of the operation of PSHPs in a spot-market relies heavily on the relative difference between the electricity prices in off-peak hours (when the unit buys energy to pump the water from the lower to the upper reservoir), and in peak hours (when the units sell the energy generated by discharging the water from the upper to the lower reservoir), the change of the pattern of spot-market prices could have a remarkable economic impact. As a consequence, owners of PSHPs are looking for additional sources of revenues by increasing their presence in the provision of ancillary services.

The participation of PSHPs in the regulation reserve markets is not new. However, the last technical developments have been oriented to improve their capability of providing load-frequency control. In particular, special attention is given to variable speed PHES [13–15] and hydraulic short-circuit operation [5,16].

In this context, PHES “reemerges” as an excellent means to integrate wind power into the electric power system [17], not only because of its maturity and technically proven flexibility, but also because of its significant remaining potential [18–20]; and so it is demonstrated according to some recently projected values of PHES capacity in 2020 [21].

The objectives of this paper are (1) to review the current trends in the operation of PSHPs, (2) to discuss why current practices should be re-examined, (3) to check whether the existing models and approaches found in the literature are appropriate or not for an optimal planning and operation of PSHPs in the new framework, and (4) to present the main challenges faced by PSHPs owners who will need to adapt their scheduling and bidding models to optimize jointly the operation in the energy and in the ancillary services markets. It is important to note that the operation of small PHES facilities connected to local distribution grids, such as those integrated in irrigation or water supply networks, sometimes operating in coordination with small wind generators, is outside the scope of this paper. The reader interested in that type of PHES facilities is referred to [22–24].

This paper is organized as follows. Section 2 presents a review of current trends, where the discussion about their drawbacks to face the new framework is embedded in the different subsections. Section 3 presents the main challenges for the near future, and Section 4 summarizes the main conclusions of this work.

2. Trends of pumped-storage hydropower plants operation

The identified trends in the operation of PSHPs are presented in this section, divided according to four different subsections. In Section 2.1, main motivations for PSHPs operators to schedule the power generation and consumption of the PSHP considering its participation both in different energy and ancillary services markets are analyzed. Additionally, some relevant multi-market PSHP scheduling models are described. Section 2.2 focuses on how PSHPs operators cope with the different sources of uncertainty existing in liberalized market contexts, namely: the energy and ancillary services prices, the behavior of competitors in the diverse markets, the real time use of committed power reserves. Some discussions are included about the effect of the uncertainty on the fulfilment of longer term operation guidelines, such as target stored volumes at the end of the day or the week. In Section 2.3, the coordinated operation of wind and pumped-storage power plants is discussed, from the perspective of both the TSO and the PSHP operator. Section 2.4 deals with the operation of PSHPs capable to provide load-frequency control in pumping, or consumption mode, by means of variable speed or hydraulic short-circuit operation. Finally, in Section 2.5 main observed trends are summarized. Several comments have been included across Sections 2.1–2.4 in order to define the power scheduling models used in the cited papers, putting special emphasis on the optimization techniques used in the model, and other model features which are summarized in a table in Section 2.5. Additionally, some comments on which are the most widely used optimization techniques for short- and long-term power generation scheduling of PSHPs are included in different subsections.

2.1. Multi-market operation

Since the beginning of the nineties, power systems all over the world have gradually experienced a continuous process of deregulation. These new market schemes have served as a framework for the development of plenty of planning tools that assist the power generation companies to better utilize their energy resources.

As a result of the above-mentioned deregulation process, a wide number of electricity markets schemes have appeared all over the world. In spite of the differences among the distinct designs and rules of existing electricity markets, the activities of buying and selling electric power are in most cases organized around a short-term wholesale energy market. These markets are supervised by an independent agent usually referred to as the Market Operator, who is in charge of the market clearing process,

from the selling and buying bids submitted respectively by the generation companies and consumers. As a result of the market clearing process, the Market Operator determines the energy prices as well as the energy to be delivered by every power generation company, power plant or group of power plants, at each programming period of the time horizon, usually 1 day.

In order to allow the agents correcting any deviation in their generation or consumption forecast due to, for example, unexpected unit unavailabilities or weather conditions, different successive markets, usually referred to as intraday markets [25], take place every day once that the spot market has been cleared.

Besides the day-ahead and intraday markets, there exist other markets wherein certain ancillary services that contribute to guaranteeing the quality, reliability and security of supply, are negotiated; frequency and voltage control are examples of said services [26,27]. The amount of ancillary services and consequently, the economic volume negotiated in the corresponding markets, have considerably increased during recent years [28], due to the increase in the penetration of non-dispatchable RES in many electric power systems.

Most of the power generation scheduling models found in the technical literature have been developed to be used within a specific electricity market and for a specific generation technology, or a mix of these. Nevertheless, most of the approaches proposed in the literature can be easily applied to a wide range of electricity market schemes, as well as to different power plants or systems. For this reason, authors trust that the reader will understand that some discussions on thermal and hydrothermal scheduling models have been included in the paper, as a means to introduce some important issues for defining or planning the operation strategy of a PSHP.

In view of the complex structure of the deregulated electricity markets, it seems logical that there has been an increase in the number of markets or services considered in the power generation scheduling of a PSHP.

To authors' knowledge, [29] is the first paper where the importance of the share of profits that a PSHP could obtain in the ancillary service markets is emphasized. In that paper, two bidding strategies are compared in terms of the expected revenues; the former consists in bidding only for energy by using the well-known peak shaving method, whereas the latter consists in bidding for both energy and ancillary services, by using a heuristic algorithm similar to the previous one, which compares the income the PSHP can obtain in the energy and ancillary services markets with the expected income the PSHP will obtain with the stored water, given by the so-called *water value*. The results obtained in that paper showed that a PSHP might almost double its daily income by simultaneously bidding in energy and ancillary services markets, with respect to the income obtained by bidding only in the energy market.

Some years later, [30] used a marginal approach [31] to optimally schedule the energy and reserves (both spinning and non-spinning) of a closed-loop[1] PSHP for the corresponding markets, and obtained an analytical condition depending on the energy and reserve prices, as well as on the unit cycle efficiency (generating-pumping), that should be fulfilled in order for the schedule to be profitable. Authors applied the devised strategy to schedule the energy and reserves of a single-unit closed-loop PSHP in two periods of 4 weeks, in the framework of the New York Independence System Operator (NYISO) market, and observed that within a season, the power generation and consumption schedules do not change significantly; the PSHP generates more on weekdays and pumps are more active during the weekends; both generating and pumping hours in summer are almost twice those in winter.

In [32], the above-mentioned condition was revised in order to consider the unit's start-up and shut-down costs and the head dependency of the unit's power limits and efficiency. The revised strategy was applied to schedule the energy and reserves of a single-unit closed-loop PSHP in a 1 week period, considering both daily and weekly operating cycles, within the framework of the NYISO market. The results presented in the paper show that the weekly operation mode may yield a higher profit, as well as a more efficient use of the energy storing capacity. The revised strategy was compared, in terms of the expected weekly profit, with those proposed in [29,30], as well as with a traditional operation strategy (as referred to by the authors), consisting in filling the upper reservoir during off-peak hours, and emptying it during peak hours; the proposed strategy yielded 4%, 6% and 24% higher profits.

In [33] a comparison between mixed integer linear programming (MILP) and Lagrangian Relaxation (LR) abilities to solve a deterministic unit commitment (UC) problem of a power generation company is presented. The company is supposed to own a set of thermal, combined cycle, cascaded-hydro and pumped-storage units. Market prices for energy and ancillary services (spinning and non-spinning reserves) are considered deterministic. Authors assume that spinning and non-spinning reserves scheduled are always used.

In the MILP formulation, each operation mode of the pumped-storage units (generating, idle and pumping) is modeled as a pseudo-unit, as referred to by the authors. Transition costs among different operation modes as well as minimum on/off times for each mode are considered in the paper. Upper bounds for the total energy and ancillary services supplied by the company are considered in the paper. In order to cope with such a *complicating constraint*, authors propose a three-step sequential solution procedure, both for the MILP and LR formulations. At each step the problem is solved considering a different subset of power generation units (cascade-hydro, pumped-storage, and thermal and combined cycle). The above-mentioned upper bound is updated at the beginning of each step by subtracting the energy and ancillary services obtained in the previous step.

Three different cases depending on the considered type of units are studied in the paper: cascaded-hydro, pumped-storage and thermal and combined cycle. The expected profit obtained with the MILP based model is higher than the one obtained with the LR based one in the three cases. The operation schedules of the cascaded-hydro and pumped-storage units obtained with the MILP based turned out to be more responsive to market prices and made better use of limited water resources. The results of the third case show that even though it takes longer for the MILP based model to find an initial feasible solution, this solution is much closer to the optimal one than that of the LR based model. The results of the fourth case show that the computation times of both the MILP and LR based models increase almost linearly with the number of scheduling periods up to certain value beyond which that of the MILP based model increases abruptly whereas that of the LR based one still follows a linear trend.

The LR based method used to solve the UC of the pumped-storage units in [33] is described in [34]. As stated in that paper, the basic idea of the model is to relax the pond level dynamics and constraints by using a set of Lagrange multipliers. In this way, the problem can be decomposed into a series of subproblems, each corresponding to a stage or scheduling period. Assuming additionally that the operating state is known a priori, each subproblem consists in optimizing a single variable function. Once that the optimal generation or pumping level is obtained for each operating state and scheduling period, dynamic programming (DP) is used to optimize the transitions between states across hours.

[1] In this paper, closed/open-loop refers to those PSHPs which do not/do receive natural inflows in the upper reservoir.

It is interesting to note that seven different operating states are considered in [34]: normal generation, normal pumping, idle, down fringe generation, up fringe generation, down fringe pumping and up fringe pumping. The so-called fringe states are used to consider that the unit may generate or pump below minimum during the transition between normal generation, pumping, or idle.

As it is well known, the dual solution obtained from LR is not guaranteed to be feasible. In order to obtain a feasible solution from the LR results, authors propose an interesting heuristic method that consists in checking sequentially the pond level constraints and consequently modifying the generation or pumping levels in a set of hours selected as a function of their marginal cost. A similar criterion is used in [35] to guide the search within a branch and bound tree of a MINLP based short-term scheduling model of a hydropower plant.

The study presented in [34] was developed in the framework of the New England Power Pool (NEPOOL). It is interesting to note that according to the NEPOOL rules that were then in force, pumped-storage units associated to a specific reservoir must be dispatched in such a way that the reserve contribution is maximized.

As it has been emphasized first in [29], and later in [36,37], among others, the revenues that a PSHP can obtain from providing ancillary services may be of a pretty considerable magnitude in comparison with those that can obtain from just levelizing the power demand. It is worthy to mention in this sense the works presented in [38,39]. In the former, a MILP based deterministic model is used to calculate the bids of a price-taker PSHP for the day-ahead and spinning reserve markets; bids consist in a single hourly value for each market. The hourly prices of the day-ahead and spinning reserve market, as well as those corresponding to both the up- and down-regulation energy requested in real time by the TSO, are modeled by means of a single scenario. A fixed percentage of the committed spinning reserve is assumed to be requested in real time by the TSO. Three different cases were analyzed in the paper in terms of the share of wind generation in the power demand supply; the bigger the share of wind, the lower the day-ahead market prices and the higher those of the spinning reserve market.

Even though the MILP based model presented in [38] is somewhat simplified and that the design parameters of the PSHP used in the case study are somewhat particular, some results presented in the paper are worthy of mentioning. Such is the case of the fact that in the three above-mentioned cases, the optimal operation strategy leads to negative revenue in the day-ahead market. Considering only the so-called base case, in which the hourly prices of the day-ahead and reserve markets, as well as the power delivery requests in the latter, are historical, the revenue obtained in the day-ahead market is negative, whereas that obtained in the spinning reserve market is positive and more than one order of magnitude bigger than the previous one, in absolute value. According to the results, the expected share of revenue obtained in the spinning reserve market could even increase with respect to the base case as the share of wind generation increases, under the assumptions used in the paper to predict the impact of such increase in the share of wind on the day-ahead and spinning reserve markets prices.

In [39], authors compare different price arbitrage strategies in a series of energy-only electricity markets where no ancillary service markets are considered. All strategies are based on a set of operation rules and they differ from each other essentially in the future time horizon for which the hourly energy prices are assumed to be known. The operation of a PSHP with a 6-h charge/discharge cycle, following each strategy, is simulated during a 1-year period, using data from different electricity markets. The results of the simulations show that the expected profit of the PSHP differ considerably from one electricity market to another; the maximum expected profit ranges from 20 to 23 times the minimum one.

From the results of the above-mentioned simulations, the authors determined the most profitable strategy and simulated the operation of the PSHP, in accordance with the said strategy, during a 5-year period (2005–2009), using data from different electricity markets (UK, Alberta, Italy, Nordpool, Portugal, and New Hampshire). The obtained results show that even for the same electricity market, the expected profit may vary considerably from 1 year to another. In order to gain insight on the economic feasibility of the PSHP, the expected profits were compared to the annual repayments corresponding to different values of the interest rate and the initial investment cost. The results show that even with a low investment cost and a low interest rate, a PSHP is a risky investment in most electricity markets, as long as the operation strategy is based on the price arbitrage.

The results obtained in [39] give full evidence that the operation strategies based on the traditional price arbitrage are far from being economically feasible for a PSHP in a liberalized market context, and provide a sound numerical support to the conclusions drawn in [29,37], regarding the importance of making a profit from the reserve markets.

The existing difficulties to make profit from the traditional price arbitrage in a large number of electricity markets all over the world, along with the environmental concerns, are the main limiting factors for a further development of PHES, and the reason why, as observed in [40], investors are showing more interest in upgrading or repowering existing PSHP or conventional hydropower plants. Consistently with this trend, recently there have been several efforts to identify the most suitable existing hydropower and non-hydropower reservoirs for integrating PHES as can be seen in [41,42].

As stated in [40], increased wind power penetration might contribute to increasing difference between peak and off-peak prices, and thus increase the profitability of PHES from the traditional price arbitrage. Nevertheless, as shown in [43], wind power may or may not contribute to increasing the number of hours in which the generating-pumping cycle can be profitable, as a function of its covariance with the load demand, or with the energy prices in a market context.

Apart from considering the participation in different markets, the strategy of the PSHP owner can vary substantially depending on the market rules. For instance, in systems where the market clearing algorithm is based on simple bids (pairs *quantity-price* submitted independently for each hour), the responsibility for obtaining feasible schedules is transferred to the market participant. By contrast, markets based on complex bids are able to capture many of the existing technical constraints like for instance the maximum capacity of the upper or lower reservoir.

The importance of building offer curves or bids for hydro plants is emphasized in [44]. Even though no PSHP is considered in the paper, it is worthy to describe it briefly. In [44], a multi-stage stochastic MILP based model is proposed to calculate the bids of a price-taker power company for the day-ahead energy and reserve markets. The company is supposed to own a set of hydropower plants located in a river catchment. A periodic autoregressive process is used to generate both the vectors of electricity prices used to build the scenario tree and those used to evaluate the expected profit corresponding to the bidding plans obtained as a result of the model.

The operation schedule obtained with the proposed model is compared with another one obtained as a result of an equivalent model in which the bid prices are not considered as decision variables. The results presented in the paper show that both the operation schedule of the hydro units and the expected revenue

can vary substantially depending on whether or not the bid prices are considered as decision variables; actually, in the example analyzed in the paper, the expected profit corresponding to the model where bid prices are considered as decision variables turns out to be a 13.4% higher.

2.2. Modeling and management of market uncertainties

As hinted at both in [29,30], the uncertainty in the market clearing prices may significantly influence the profits made from both energy and ancillary services markets. As it is well known, electricity prices present a very high short-term volatility [45]. Consistently, during the last decade, several approaches were proposed to cope with the uncertainty associated both to energy and ancillary services prices, in the power generation scheduling and bidding problems.

In outline, it can be stated without loss of generality, that there are two different approaches to face the above-mentioned uncertainty: risk-neutral and risk-averse. In the risk-neutral approach, the decision maker tries to optimize the expected value (for instance, the expected profit). By contrast, the risk-averse approach aims at quantifying and to limit the exposure to risk [46] of the power generation schedule or bids; risk-averse power generation or bidding models are usually referred to as "risk-constrained" [47].

The approach proposed in [30] rests on the perfect knowledge of the weekly market clearing prices, which is not possible. In order to consider prices uncertainty, authors also proposed a somewhat naive scheduling strategy, based on historical prices, which could be understood as a risk-neutral way to cope with prices uncertainty. The results presented in the paper show that the profit obtained when following the naive strategy are only from 3% to 6% lower than those obtained assuming a perfect knowledge of the market clearing prices.

One of the first papers where a risk-averse approach was used to cope with prices uncertainty in the short-term power generation scheduling of a PSHP is [48]. In that paper, a stochastic MILP based formulation is used to derive the energy and reserve offer curves of a hydrothermal power system with pumped-storage units. The particular characteristics of both energy and reserve offer curves are considered in the model, according to the regulations of the electricity market where the model is intended to be used (New England market). The problem formulated by the authors considers certain own load and reserve supply obligations or self-scheduling requirements, as referred to by the authors. These self-scheduling requirements couple to each other the decisions of each single generating unit and thus constitute a sort of complicating constraints [49]. Authors use a LR based approach in order to cope with these complicating constraints in an iterative manner.

The bidding risk in [48] is considered by including a weighted penalty term in the objective function as a function of the energy and reserve prices variances. By varying the weight value, a set of trade-off solutions can be obtained for which the expected profit cannot increase without increasing risk; the so-called *efficient frontier* [50].

The use of the variance as a risk measure has been criticized for its symmetry with respect to the expected value of the revenue, what implies that certain revenues above the expected one are considered as risky values [47]; however, some authors have recently used it to consider the risk in pumped hydro scheduling problems [51].

Other risk measures, such as the value at risk (VaR) or the conditional value at risk (CVaR), are considered more efficient than the variance as a risk measure; the latter has been widely used for hedging against risk for power generation scheduling or bidding purposes [47]. In [52], the CVaR is used as a risk measure within a day-ahead power generation scheduling problem of a hydrothermal system with pumped-storage units. The problem is formulated as a linear programming (LP) problem. A discrete set of hourly energy prices scenarios is considered in the paper. The objective function of the problem consists in minimizing the expected economic loss; i.e. fuel cost of thermal generating units plus cost of energy purchased for pumping minus revenues from the energy sold in the day-ahead market, physical agreements and contract differences. No reserve or ancillary market is considered in the paper. Authors include a specific constraint in the problem in order to limit the maximum CVaR of the economic loss as a function of the expected loss, and obtain the profit-CVaR efficient frontier by varying the percentage of expected loss set as maximum CVaR value.

One of the most immediate methods to hedge against the risk of prices uncertainty is imposing a minimum profit (or maximum loss) for every considered prices scenario. In [53], the use of a minimum profit constraint was compared with the use of a specific constraint on CVaR value in the day-ahead power generation scheduling problem of a set of hydropower plants. The expected profit with the minimum profit constraint turned out to be lower than the one obtained with no risk-aversion constraints, and this in turn, was lower than the one obtained by imposing a constraint on the CVaR value. Even though these results may vary as a function of right-hand side values of both minimum profit and CVaR constraints, it is worthy to take a look at the histograms of the profit obtained for each scenario shown in the paper, for both the minimum profit and CVaR approaches.

An alternative method to reduce the exposure to risk, which has been commonly used for power generation scheduling and bidding purposes, consists in limiting or minimizing the so-called *downside risk*; i.e. the failure to meet a target minimum profit [54]. As in the case of using a minimum profit constraint, both power generation schedule and expected profit can be significantly sensitive to the selected minimum downside, even resulting in infeasible solutions when very ambitious values are used [55]. In [54], a stochastic MILP based model is proposed for the midterm generation scheduling of a hydrothermal power system, with pumped-storage units. The objective function of the model is aimed at maximizing the profit for participating in both the energy and reserve markets. The energy and reserve markets prices, as well as the generators forced outage and natural water inflows in hydropower reservoirs are considered in the paper by means of a scenario approach. A specific constraint to limit the maximum expected downside risk is included in the model. The time horizon of the model is 52 weeks. The model is solved by means of a LR based approach. Authors analyze among other things the influence of the time step on the expected profit, by comparing the results obtained with the model when using a fixed 1-h time step, and a variable time step (the first month with hourly steps, the second and third months with 2-h steps, the second season with 4-h steps, the third season with 6-h steps, and the remaining months with 8-h steps). As stated by the authors, using the variable time step approach yields good scheduling solutions which satisfy the most important constraints with a reasonable computation times.

All the above-mentioned scheduling models were developed for a price taker producer. This is usually the case in electricity markets with a high degree of competition, where the size of every company is considerably small with respect to the size of the power system. In [56], a stochastic MILP based model is used to prepare the *portfolio bids* of a price-maker producer for the day-ahead, intraday and secondary regulation reserve markets of the Spanish power system; the company is assumed to own thermal, hydro and pumped hydro generation assets. In order to consider the influence of the company supply bids on the day-ahead market

clearing price, a series of *residual demand curves* (RDCs) were considered for each hour of the day. RDCs are also called *price-quota curves* [57] and they represent the market clearing prices for different production levels (quotas) of the company. The RDC faced by company can be obtained by subtracting the aggregation of all competitors' supply bids from the demand-side's buying bids [54]. Both the RDC and the revenue function of a generation company can be estimated from historical market data, provided that information of submitted bids are publicly available after a certain time, as it is the case of the Iberian electricity market. The revenue function gives the revenues corresponding to different values of energy offered. Integer variables are used in [56] to deal with the nonconcavity of the revenue functions and to impose that the energy offered in the supply bids must increase as the market clearing price increases. The uncertainty associated to the intraday and reserve markets' prices, as well as the influence of the company energy and reserve bids on the results of these markets, were not considered in that paper. The model is solved by means of a Benders decomposition (BD) approach.

The influence of the energy or reserve bids of a power plant or company in the market clearing prices increases as the relative size of its bids increases, with respect to the total volume of energy or reserve negotiated in the corresponding market [58]. It seems therefore logical that bids presented in balancing or reserve markets, such as the Nord Pool's balancing market (Elbas), or the secondary regulation reserve market in the Spanish power system, are more likely to have an impact on the corresponding market clearing prices, due to the relatively small volume of reserves negotiated in this type of markets.

In [59] a multi-stage stochastic MILP based model is proposed to calculate the bids of a price-maker power company for the day-ahead, the secondary reserve and the first intraday markets. The power company is assumed to own thermal, hydro and pumped-storage units. It is important to note that authors assume that start-up decisions of thermal units are taken as input from the results of a longer-term model. The influence of the power company on the prices of the above-mentioned markets is considered through a set of scenarios of RDC. In a similar way to [56], the RDC patterns are piecewise linearized, and thus the hourly income corresponding to each RDC, is given by a quadratic concave function of the total hourly energy or power offered by the company in the corresponding market. The said function is in turn approximated in a clever manner by means of a set of linear tangent cuts, and thus can be used within the MILP based model. In contrast to [56], in [59] authors obtain the hourly offer curve of each generating unit, for each considered energy and reserve market. Particular characteristics of each market bids are fully considered in the paper, as suggested in [44].

Pumped-hydro units are modeled in a somewhat simplified manner, by means of a single energy balance equation; no decisions on the operation mode (generating/pumping) or the on/off status of the units are considered. Previous formulation was improved in [37], where binary variables were used to select the operation mode as well as the on/off status of the units.

It is worthy to mention that a significant change in the strategy of the pumped-hydro portfolio was observed in [37], as a consequence of considering the intraday market when bidding in the day-ahead and secondary reserve markets. As shown in the paper, the intraday market might be relevant for solving possible infeasibilities that could appear after the clearing of the reserve market, whose consideration yields in a deterministic case a 5% increase in the daily revenue, in agreement to some extent with the conclusion drawn in [29] about the high share of profits expected to be obtained by a PSHP from ancillary services markets.

An original approach to model the influence of the reserve bids of a hydrothermal system with pumped-storage units on the clearing prices of RWE Net AG and E.ON Netz GmbH tertiary reserve markets can be found in [36], where a stochastic non-linear programming (NLP) based model is used for simultaneously bidding in the European Energy Exchange spot market (EEX) and the above-mentioned reserve markets. The behavior of competitors in the reserve markets is modeled through a joint probability distribution of the market clearing prices, as a function of the producer's reserve bidding capacity. As stated by the authors, the results presented in the paper show that the expected daily profit is “dominated by the reserve markets products”, in agreement with [29,37].

According to the above discussions, everything seems to indicate that there should be a shift from the traditional price arbitrage based operation to a “reserve driven operation”, for a PSHP to be profitable. Nevertheless, in order to maximize the profits that a PSHP can obtain from the reserve markets, there is a task that to authors' knowledge remains challenging, and which can be of crucial importance in quite a few electricity markets, particularly for closed-loop PSHP: considering the uncertainty associated to the amount of effective power reserve requested in real time for load-frequency control purposes.

As it is well known, each electricity market has its own structure and market mechanisms for dispatching or assigning power regulation reserves [60]. In a significant number of reserve markets, such as in Spain [61], Belgium [62], Ontario [44] and France [27], among others, power regulation or balancing reserves are remunerated for two different concepts, power reserve availability and delivery, the latter depending on real time contingencies, power and load fluctuations [28].

To authors' knowledge, the first attempt to consider the uncertainty associated to the real time use of a PSHP committed power reserves was done in [51]. In that paper, a MILP based model is proposed to maximize the weekly profit of a PSHP in a multi-market framework. The uncertainty associated to the hourly prices of the energy, spinning reserve and regulation markets, is considered through the variance of the corresponding forecast errors. The uncertainty associated to the hourly power delivery request in the ancillary services markets is modeled through a suitable probability parameter, which is explicitly included in the water balance equation of the upper reservoir. Particular characteristics of each market bids are not considered in the paper, but rather a single hourly value for each market is obtained as a result of the model, in a similar way to [63]. It is interesting to note that the operation and maintenance costs are calculated as a function of the powers scheduled for the energy and ancillary services markets; the latter are therefore weighted by the corresponding probability parameters.

Several risk terms are included in the objective function of the problem, each weighted by a risk penalty factor. In a similar way to [48], these factors are to be specified by the user according to his/her degree of risk aversion. The influence of each risk penalty factor on the expected profit was analyzed separately. It is worth noting that the risk penalty factor corresponding to the spinning reserve prices turned out to have by far the most significant impact on the expected profit, mainly due to the high percentage of the profit that is expected to be obtained from the spinning reserve market.

In [64], authors present a model similar to the one presented in [51], which is formulated as a mixed integer nonlinear programming problem, in order to accurately model the so-called head effects on the hydro generation characteristic. The objective function of the model is aimed at maximizing the daily profit of a set of cascaded hydropower plants and one PSHP. The model is solved with the SBB solver [65].

Since usually, the rated head of closed-loop PSHPs is considerably large, head effects are of minor importance for such type of

PSHPs, and therefore, head is normally considered constant throughout the time horizon as in [66]. This is not necessarily the case in open loop PSHPs, where the rated head is strongly influenced by other parameters different from the economic feasibility of the generating–pumping cycle. In this sense, it is worthy to mention [67], where a MILP based model is proposed to calculate the short-term operation schedule of an open-loop multiunit PSHP. The formulation proposed to model the hydropower generation characteristic is similar to the one proposed in [68], where three piecewise linear non-concave power-discharge curves were used to model the hydropower generation characteristic, each corresponding to a different reservoir level. A smart MILP formulation was proposed to select one of the curves as a function of the actual reservoir level; each curve was used to compute the plant power output within a specific interval of reservoir volumes. In [67], a reduced set of power-discharge curves is also used to model the hydropower generation characteristic. Nevertheless, the curves are convex, what contributes to reducing the number of binary variables and therefore the CPU time, and the power output is obtained by interpolating between the two "nearest" power-discharge curves, what yields a more accurate power estimation.

In [69], the influence of the so-called power delivery request on the expected revenues of a PSHP is considered with certain rigor. In that paper, a mixed integer nonlinear programming (MINLP) based model is proposed to obtain the power offers of a wind power plant and a PSHP for the day-ahead, spinning and regulating reserve markets of the following day; bids consist of a single hourly value for each market. The hourly prices of the above-mentioned markets are considered in a deterministic way, whereas the uncertainty associated to the wind power production is modeled through a set of scenarios. Each scenario is obtained as the sum of the hourly wind power forecasts and a different error value, each with a different probability that is, in turn, estimated from historical data. A neural network is used to calculate the hourly wind power forecasts. Different probabilities of power delivery request are considered in the paper.

An interesting comparison between uncoordinated and coordinated operation is presented in the paper. The results included in the paper show that a significant increase in revenues could be obtained as a result of the coordinated operation. As it is shown in the paper, the higher the probability of power delivery request in both the spinning and regulation reserve markets, the bigger the added value obtained as a result of the coordinated operation.

In [70] (see Section 2.4), authors implicitly consider the uncertainty in the real time use of committed regulation reserves by using a 24-h rolling horizon with a 5-min time step, which is updated, along with the reserve requirement, as new wind forecasts become available, in a similar way to [71].

The real time use of committed power reserves does not only affect the PSHP current day or week income, but also the fulfilment of the end of day or week target stored energy, and thus the next day or week income. Traditionally, the target value is determined in such a way that the PSHP can make as much profit as possible from the price arbitrage. For daily-cycle PSHPs, this usually means to begin and end everyday (0:00am) with the upper reservoir empty. For weekly-cycle PSHPs, the end of week target strongly depends on the difference between weekend and Monday energy prices.

Except for one of the cases analyzed in [39], in all the above-discussed papers either an end of day or week target stored energy [51], or an end of day or week marginal *water value* function [37] is assumed to be known in advance. The use of one or another approach is not an arbitrary decision, but rather it depends, among other things, on the characteristics of the PSHP under study. For closed-loop PSHPs, a target value is normally used.

Noteworthy results were obtained in [39] regarding the determination of the end of day targets of a closed-loop PSHP in a liberalized market context. In that paper, the profit obtained assuming a perfect forecast of the hourly energy prices over a 1-year period is used as a basis for comparing different short-term price arbitrage strategies. It is interesting to note that, according to the results of the 1-year period simulations, the most profitable strategy would allow a PSHP with a 6-h charge/discharge cycle (daily cycle) to obtain a 97% of the profit obtained with perfect forecast; however, when considering a 24-h charge/discharge cycle (weekly-cycle), the most profitable strategy yields an 82% of the maximum theoretical profit. Taking into account that the planning horizon of the most profitable strategy is 1 day, these results demonstrate that a daily-cycle closed-loop PSHP does not need an accurate price forecast for a time horizon longer than 1 day and equally important, that the intraday price arbitrage strategy may be far from the optimum for a weekly-cycle closed-loop PSHP.

These results shall be understood in the context of a price arbitrage strategy, what according to other results presented in [39] does not allow justifying the investment in a new PHSP. Considering the ongoing shift in the operation of PSHPs from the traditional price arbitrage to an operation more focused on the ancillary services markets, further studies on the determination of the end of day and week targets of closed-loop PSHPs are necessary.

Even though in a centralized market context, this topic has been recently analyzed in [72]. The objective of that paper is to establish day-ahead and week-ahead reservoir targets for a PSHP with a large storage capacity in an electric power system with high wind power penetration, from the TSO point of view. The main hypothesis of the study is that a fixed charge/discharge cycle is blind to the future wind power profile and that, as a consequence, does not allow exploiting the storage capacity to its full potential.

In order to fully exploit the PSHP storage capacity, authors propose a two-step optimization procedure. The first step consists in calculating the power generation scheduling of all generating units of the power system by means of a MILP based stochastic model that uses a set of annual historical scenarios of hourly wind power production, all of them with the same probability; the model could be easily adapted to consider also the hourly power demand as a stochastic variable. The time horizon of the model is 1 year, divided into 6-h intervals. The objective function consists in minimizing the power supply costs. Weekly storage targets are derived from the reservoir weekly end levels obtained as a result of the model. Non-anticipativity is considered in the model, as a soft constraint (i.e. through a penalty term in the objective function), in order to force the reservoir weekly end levels to be equal across all scenarios.

The second step is aimed at deriving daily storage targets, even though it can be understood as a procedure for daily power generation scheduling. A MILP based model, analogous to the above-described one, is used here with a time horizon of 1 week, divided into 2-h periods. The model is run day by day; i.e. only the results corresponding to the first day of the week are considered as a firm operation schedule. For each run, a set of 7-days wind power forecasts is used. The reservoir level at the end of each day is forced to be equal across all scenarios. In addition, the reservoir level at the end of the week is forced to be equal to the end of week target previously obtained across all scenarios.

The two-step procedure proposed in the paper is compared on an annual basis to a *weekly refill* method that consists in forcing the reservoir to be full at the end of each week; end of day storage targets of the weekly refill method are set as the average of the values obtained for each day ahead wind power forecast. Additionally, a perfect foresight case was used as a reference for comparison.

The results show that the two-step procedure proposed in the paper provides a lower increase in cost, relative to the perfect foresight case, than the weekly refill method. Also, results show that the two-step procedure allows a greater flexibility in the PSHP operation, increases the PSHP capacity factor and reduces wind curtailments.

For open-loop PSHPs, both weekly storage targets and marginal water value functions have been recently used in the literature [37,67]. These targets or water value functions are usually obtained as a result of a long-term scheduling problem. The most widely used techniques for long term hydropower generation scheduling are stochastic dynamic programming (SDP) [73] and stochastic dual dynamic programming (SDDP) [74]. The use of one or another technique depends mainly on the "size" of the hydro system under study. The former is subject to the so-called curse of dimensionality [75] and therefore, is limited to reduced size hydro systems with one or a few reservoirs [76], unless aggregate or composite system representations are used [77]. The latter combines the ability of SDP to solve multi-stage stochastic programming problems, with the use of Benders cuts to approximate the marginal water value functions, and has been successfully used for long-term generation scheduling of large size hydro and hydrothermal systems during the past two decades [78].

As in the short-term range, electricity markets considerations have been gradually introduced in long-term hydro scheduling models during past decades [79]. Nevertheless, given the long time horizon of these models, and with that, their generally large computational requirements, the introduction of electricity markets considerations in the long-term range has occurred at a slower pace, and still today remains a challenging task.

To authors' knowledge, the long-term hydro scheduling model presented in [80] is the one where uncertainty in hourly energy prices has been considered with a greater degree of detail. The model presented in that paper is based on SDDP and considers the bidding decisions in the day-ahead and intraday markets within the EEX framework. The time horizon of the model is 1 year, divided into daily steps, and these in turn into hourly stages, according to the programming periods of the EEX [81]. The model was tested on a hydropower system located in the Austrian Alps, considering six different configurations that depend on the number of reservoirs and maximum and minimum generation and pumping capacities.

Another noteworthy long-term generation scheduling model is the one presented in [82]. In that paper, a SDDP-based model is proposed for the long-term scheduling of a hydrothermal system with pumped-storage and wind power. Even though the uncertainty in energy prices is not modeled with the same degree of detail as in [80], in terms of both the number of markets considered and the time resolution, it is fair to highlight that as far as we know [82] is the first work where the start-up costs of pumps and thermal generating units are considered within a long-term generation scheduling model; the method used for this purpose was first proposed in [83] for thermal generating units only.

2.3. Wind-hydro coordination

The coordinated operation of wind and conventional or pumped hydro power has been subject of study during, at least, the last 10 years [84]. It is possible to classify these studies according to the scale in which the coordination is taken place: (1) a system-based coordination approach, where the coordination is aimed at supporting the system to reduce, among other things, the power supply costs and wind curtailments, or (2) a local coordination approach, where the coordination is aimed at maximizing the profit of the hybrid power plant owner.

In [85], authors analyze the impact of wind power and PHES in the generation mix and net load profile of the Irish power system, and assess the revenue obtained from the joint operation of wind and PHES as well as the storage capacity which best exploits the coordination between wind and PHES considering three different heuristic operation strategies. The results presented in the paper show that PHES may help to significantly reduce the use of peaking plants, as well as to integrate a larger amount of wind power in the electric power system.

In [86], authors analyze the contribution of a PSHP to reducing the power supply costs and curtailed wind power of a hydrothermal power system with a large wind power penetration level, as well as its effects on the generation mix of the system and the number of start-ups of the thermal generating units. For this purpose, authors use both deterministic and two-stage stochastic UC models based on MILP. The objective function of the models is aimed at minimizing the power supply costs, including the start-up costs of the thermal generating units. The time horizon of the UC models is 48 h, and is divided into hourly periods. A total of 10 scenarios of 11-month length are sequentially processed with the UC models. The uncertainty associated to the load demand, available generation capacities and wind power is explicitly considered in the stochastic models by means of a scenario approach. Even though the paper is more focused on comparing different approaches to cope with the uncertainty in the UC problem, from the results presented in the paper, authors conclude that the flexibility provided by the PSHP is highly valuable in the power system under study, especially in terms of power supply cost savings.

In [71] the benefits of a PSHP on the operation of the Irish electric power system are analyzed considering different levels of wind power penetration. The paper focuses on the contribution of the PSHP to reducing wind curtailments, system operation costs and carbon dioxide emissions, as well as on the shifts in the PSHP operation with respect to the traditional price arbitrage based operation, and on the influence of the PSHP on the electrical energy imports and exports between Ireland and Great Britain.

For these purposes, authors use a stochastic MILP based UC model [87] with hourly time resolution and a 36-h rolling planning horizon, updated every 3 h as new wind and load forecasts become available; i.e. here-and-now decisions are made for the following 3 h. Wind power uncertainty is considered by including multiple scenarios of future wind power production.

The objective function of the model is aimed at minimizing the costs for meeting the system demand and primary and replacement reserves. Specific constraints are included in the model to prevent simultaneously pumping and generating, consistently with the use of reversible pump-turbines. The PSHP is assumed to be able to offer spinning reserve in both operating modes; in pumping mode, it is assumed that the pump can be shut-down in order to reduce power demand. The value of energy stored in the upper reservoir at the end of the day is determined by the marginal cost of water storage at the end of the previous day.

The model is run sequentially for a 1-year period, with and without a PSHP (500 MW of installed capacity and 5 GWh of energy storage), and considering several cases of wind power installed capacity (6 GW, 7.5 GW, 9 GW, 10.5 GW and 12 GW). Maximum and minimum power demands considered in the study were 9.6 and 3.5 GW, respectively.

The results presented in the paper indicate that the investment in the proposed pumped storage hydropower plant could be justified from a wind installed capacity greater or equal than 8.5–9 GW (48–51% of energy from wind), assuming a PSHP lifetime of 40 years. It is interesting to note that the wind penetration level beyond which the investment in the PSHP begins to be justified turned out to not to be very sensitive to the considered discount rate and lifetime considered.

With respect to the PSHP operation, results show that for a wind installed capacity higher than 6 GW (34% of energy from wind), there would be more days in which the PHSP would pump at peak hours and generate at off-peak ones, as wind energy would be more frequently the marginal technology to meet the system demand. As it seems logical, results show that as wind penetration increases, Ireland would go from being a net energy importer to becoming a net energy exporter; nevertheless, energy exports would decrease with storage since the system could either benefit from the lower energy prices of Great Britain or store the excess wind power instead of exporting it.

Finally, it is worthy to highlight that as demonstrated in [71], it is important to analyze the impact of PSHP taking into account wind stochasticity since, otherwise, wind curtailments would be underestimated and, as stated by the authors, the more uncertain the wind energy is, the more beneficial is the flexibility of the PSHP.

In [88] a deterministic MILP based minimization-cost model is proposed to calculate the optimal short-term operation schedule of the world's first Megawatt-level renewable energy system in El Hierro Island (Spain). The system is composed by five wind turbines, each of 2.3 MW, and a pumped storage hydropower system comprising four Pelton turbines (2.83 MW each) and eight pumps (six fixed speed 0.5 MW pumps and two variable speed 1.5 MW pumps, which are able to regulate power consumption between 1 and 1.5 MW each). In addition, there are some thermal units to be used only in case of an emergency.

The objective function of the model consists in minimizing the sum of thermal generation costs (fuel and start-up costs) and six "artificial" cost terms related with the energy stored in the upper reservoir and wind power curtailments, among others. Coefficients of the artificial cost terms are defined by using a heuristic criterion based on the regulations in force at that time.

Since the pumped storage hydropower system comprises two different pipes (one for pumping water flow and the other one for water discharged flow), the scheduling model considers the possibility of simultaneously using both pumps and turbines. Given the high wind penetration in the system (maximum historical power demand is 7.8 MW), both inertia and spinning reserve requirements become critical for power generation scheduling, and therefore, are considered as constraints in the scheduling model. It is important to highlight that to authors' knowledge, this is together with [71], one of the few works where inertia requirements are explicitly considered within a power generation scheduling model, consistently with the special characteristics of the power system under study. In order to emphasize the importance of including the above-mentioned requirements on both spinning reserve and inertia, authors compare the power generation schedules obtained with and without considering such requirements for a specific scenario. As expected, the power generation schedules turned out to be pretty different from each other, the most noteworthy difference being the continuous simultaneous use of both pumps and turbines to minimize wind power curtailments and meet the inertia and reserve requirements.

Not only the coordinated operation of wind-hydro systems has received interest in the literature, but also the optimal sizing of such hybrid power generation systems. In [89,90], the optimal sizing of a wind-hydro system in two different island power systems is addressed by using diverse simulation algorithms based on heuristic operation rules. In the former, the algorithm is aimed at maximizing the exploitation of a predetermined volume of the water storage subsystem, while guaranteeing that the energy is supplied at minimum cost. In the latter, the algorithm is aimed at maximizing the net present value of the investment. A similar study can be found in [91], in the context of continental power systems. In that paper, authors study the optimal sizing of a wind-hydro power generation system in six different sites in the Turkish Marmara region. In a similar way to [89,90], authors use a simulation algorithm based on heuristic operation rules, aimed at meeting electricity demand at minimum cost, while avoiding to the possible extent the import of electrical power from the national electric power company.

Under the category of research works that study the benefits of a joint operation between a wind farm and a PSHP under a local approach, [63] presents a MILP based model used to calculate the power bids of a wind farm and a PSHP for the day-ahead market considering the uncertainty associated to both the hourly prices and wind power, by means of a suitable scenario tree; in a similar way to [69], bids consist of a single hourly power value. In addition to assessing the benefits that could be obtained as a result of the coordinated operation, authors analyze the evolution of said benefits as a function of the installed power of the PSHP. The results presented in the paper show that the benefits of the coordinated operation increase as the PSHP size increases; nevertheless, the rate of increase decreases as the PSHP size approaches that of the wind farm.

In [92], three different MILP based models were proposed to calculate the power generation schedule of a wind farm and a PSHP, with a 1-day time horizon and 15-min time steps. The main contribution of [92] is the consideration of the intrahour variations of wind speed. Two of the models deal with the forecast errors of wind power output by means of a chance-constrained and a scenario-based approaches, whereas one of the models is deterministic. In a similar way to [63], penalties for deviation of aggregate output power with respect to the scheduled aggregate output power are considered in the paper. A similar approach to [92] was proposed in [93] to calculate the power bids of a wind farm and a set of cascaded hydropower plants, without pumped-storage, for the day-ahead electricity market. As in [63], the uncertainty associated to wind power and hourly energy prices is considered via a scenario approach. The main contribution of [93] is, as in [92], the consideration of the intrahour variations of wind speed. Wind speed scenarios are divided into 10 min intervals, in contrast to the traditional approach based on the hourly average values. Wind speed average values are modeled as a first order Markov chain, whereas wind speed forecast errors are represented by an ARMA model. Two different cases regarding the look-ahead scheduling period (3 h and 1 day) are used to analyze the pros and cons of coordinated operation, the convenience of considering a deterministic or a stochastic approach, the influence of the imbalance prices, or rather penalties, on the power generation schedule, and the convenience of incorporating a risk aversion factor similar to the one presented in [55], as suggested in [63]. The results obtained in the paper show that the imbalance prices do have a strong influence on the benefits that the coordinated operation may yield. So much so that for high imbalance prices, the payoff corresponding to the uncoordinated operation may be larger than the one yielded by the coordinated operation. Nevertheless, from the point of view of the electric power system, the coordinated operation always results in lower wind curtailments, regardless of the use of a deterministic or a stochastic approach.

From the literature, it seems clear that the coordinated operation of wind power and PHES may yield significant benefits both for the owners of such hybrid power plants and for the power system. However, it is not that clear that these benefits can be translated into the necessary income to make the investment profitable. In this regard, it is worthy to mention the work presented in [94] where authors study the profitability of new potential wind-hydro schemes both in the Norwegian and German power systems. For this purpose, authors formulate a stochastic

optimal control problem which is solved by means of DP. The objective function of the problem is aimed at maximizing the net present income of a hypothetical investor. The planning period is 30-year length and is divided into annual steps. The decision variables of the problem are the numbers of new wind and wind-hydro power plants to be built each year. Scenarios are generated by means of Monte Carlo simulations. The problem is solved for different levels of price premium and subsidies on the investment costs. The results show that under the considered assumptions, investing in a wind-hydro power plant without public support is not profitable in any of the power systems, compared to only investing in a wind power plant. Actually, only with pretty high price premiums and/or subsidies the investment may become profitable. A similar study to the one presented in [94] can be found in [95].

2.4. Solutions for providing power regulation in consumption mode (variable speed and short-circuit operation)

The use of variable speed pumps in El Hierro wind-hydro power system is aimed at providing power regulation or balancing in pumping mode, in order to better accommodate the almost continuous wind power fluctuations of the system; it should be noted that wind installed power is higher than the peak power demand in the island. As in many other island power systems [96], El Hierro electricity market is organized in centralized way; hence the use of a minimization cost model in [88].

In liberalized electricity markets, the use of variable speed pumps may help a PSHP to make a bigger profit from the reserve markets, since it allows the PSHP to provide power regulation in both operating modes. Even though there are several PSHP with variable speed pumps or pump-turbines, both currently in operation all over the world, most of them in Japan [17], and to be commissioned soon [97], the power generation scheduling of variable speed PHSPs in liberalized electricity markets has received little attention in the technical literature.

In [98], a model based on a genetic algorithm (GA) and Tabu search (TS) is proposed to schedule the power generation and consumption of a PSHP with variable speed pumping, in order to reduce the power supply costs and increase the power supply reliability of a hydrothermal power system with a large penetration of photovoltaic power. The results presented in the paper show that the PSHP may contribute significantly to integrating photovoltaic power in the system and to improving the system reliability.

In [99] the power generation scheduling of a variable speed PSHP has been analyzed in a market context, and the contribution of variable speed to increasing the PSHP revenue in a joint energy and reserve multi-market framework has been preliminary assessed. In that paper, a MILP based deterministic model is used to calculate the power bids of a variable speed PSHP for both the day-ahead energy and secondary regulation reserve markets in the Spanish power system. Additionally, the paper compares the maximum theoretical income of a PSHP with fixed and variable speed pumping, and different flow regulation capacity in pumping mode, by sequentially executing the MILP based model during a 2-year period, using historical energy and reserve prices. Variable speed in generating mode was not considered in the paper since the PSHP under study was assumed to have a small head variation in comparison with the rated gross head.

The results presented in the paper support on the one hand the idea that the reserve market can become the main source of revenue for a PSHP, regardless of the use of fixed or variable speed, in agreement with [29,36–39,51,59], and demonstrate on the other hand, that the use of variable speed may contribute notably to increasing the revenue of a PSHP, mainly thanks to the possibility of providing load-frequency control in pumping mode.

In [70], authors assess the impact of PHES on production costs, energy prices, carbon dioxide emissions, wind curtailments and operation of thermal generating units in three different areas, considering two different levels of installed wind power. For that purpose, a commercial UC model is used [72], which is based on stochastic MILP. Nevertheless, according to [100], the study was carried out in a deterministic framework. The areas under study are the entire Western Interconnection, which covers the western part of the United States, as well as some Canadian provinces and one service area of northern Mexico; the California electricity market; and the Sacramento Municipal Utility District. A cost-minimization approach is used in the first and third areas, whereas a revenue-maximization approach was used in the second one. The time horizon of the model is 1 day, and is divided into hourly periods. The system operation in the three areas was simulated for an entire year by sequentially solving 365 day-ahead UC problems. The year under study corresponds to long-term projections for the year 2022 developed by the Western Electricity Coordinating Council of the United States. From the results presented in the paper authors state that PHES may help reduce the overall system costs, cycling of thermal generating units, transmission congestion and greenhouse gases emissions. The results show that the value of PHES increases with both wind power penetration and the use of variable speed operation.

In [101], authors assess the additional water volume that could be pumped to the upper reservoir of an existing PSHP during an entire year, as well as the corresponding additional energy that could be generated with the said water volume, thanks to the use of variable speed operation. For this purpose, authors simulate the historical daily operation of the PSHP, modifying the daily pump flows according to a higher efficiency, which is deduced from a typical Francis turbine hill curve. Even though the paper is not focused on the operation scheduling of the PSHP, it is worth mentioning that an additional energy of 5.89% might be obtained in the PSHP under study as a result of the variable speed operation, without changing its present scheduling strategy.

Besides variable speed, a special attention is currently being given to the so-called hydraulic short-circuit operation [5] as a means to provide power regulation or load-frequency control in pumping, or rather “consumption mode”. To authors' knowledge, there are only three PSHPs currently operating in short-circuit mode: Kops II [102], Malta [103] in Austria, and Geesthacht [104] in Germany; additionally, a new PSHP, specifically designed for hydraulic short-circuit operation, is currently under construction in Switzerland [16]. Nevertheless, as far as we know, the generation and reserve scheduling or bidding of a PSHP operating in hydraulic short-circuit mode has not been addressed yet in the technical literature.

2.5. Summary of observed trends

The main trends observed by the authors during the literature review carried out to prepare this paper can be summarized as follows:

Traditional operation strategies based on price arbitrage and/or peak shaving appear to be no longer economically feasible.
Ancillary services markets, particularly those related to balancing generation and demand, emerge as the main source of revenue for PSHPs in liberalized market contexts. Analogously, the support to the integration of non-dispatchable energies in the electric power system presents itself as the essential role of PSHPs in centralized market contexts, such as those currently in place in many island power systems.

Table 1
Features of the discussed papers where authors use or propose a power generation and consumption scheduling model for a PSHP or a set of PSHPs, according to the described trends.

Paper	Generation technologies	Type of PSHP	Market approach			System approach		Time horizon	Operation scheduling algorithm	Risk aversion
			Spot market	Reserve markets	Real time use of committed reserves	Regulation reserve	Real time use of committed reserves			
[29]	PSHP	CL-FS	PT (SS)	PT (SS)	FP			1 day	AHH	RN
[30]	PSHP	CL-FS	PT (SS)	PT (SS)	n/c			1 week	AHH	n/c
[32]	PSHP	OL-FS	PT (SS)	PT (SS)	n/c			1 week/1 day	AHH	n/a
[33]	HT	CL-FS	PT (SS)	PT (SS)	n/c			1 day	MILP vs. LR	n/c
[34]	HT	CL-FS				RR (SS)	n/c	1 week	LR+DP	n/c
[36]	HT	n/s	PT (MS)	PM (MS)	n/c			1 day	NLP	RN
[37]	HT	CL-FS	PM (MS)	PM (MS)	n/c			1 day	MILP	RN
[38] [b,c]	PSHP	CL-VS	PT (SS)	PT (SS)	FP			1 day	MILP	n/c
[39]	PSHP	CL-FS	PT (SS)	n/c	n/a			1 year/1 day	AHH	n/c
[43] [b]	HT	CL-VS				n/c	n/a	1 week	MILP	n/c
[48]	HT	CL-FS	PT (MS)	PT (MS)	n/c			1 day	MILP+LR	VFE
[51]	PSHP	CL-FS	PT (SS)	PT (SS)	FP			1 week/1 day	MILP	VFE
[52]	HT	CL-FS	PT (MS)	n/c	n/a			1 day	LP	CVaR
[54]	HT	CL-FS	PT (MS)	PT (MS)	n/c			52 weeks	MILP+LR	DR
[56]	HT	OL-FS	PM (MS)	PT (SS)	n/c			1 day	MILP+BD	RN
[59]	HT	CL-FS	PM (MS)	PM (MS)	n/c			1 day	MILP	RN
[63]	WH	CL-FS	PT (MS)	n/c	n/a			1 day	MILP	RN
[64]	HS	CL-FS	PT (SS)	PT (SS)	FP			1 day	MINLP	n/c
[67]	PSHP	OL-FS	PT (SS)	n/c	n/a			1 week	MILP	n/c
[69] [b]	WH	CL-FS	PT (SS)	PT (SS)	FP			1 day	MINLP	n/c
[70]	WHT	CL-FS-VS				RR (SS)	i/c (RH)	1 day	MILP	n/c
[71]	WHT	CL-FS				RR (MS)	i/c (RH)	36 h	MILP	RN
[72]	HT	CL-FS				n/c	n/a	1 year/1 week	MILP	RN
[80]	HS	OL-FS	PT (MS)	n/c	n/a			1 year	ADDP	P
[82] [e]	WHT	OL-FS	PT (MS)	n/c	n/a			169 weeks	SDDP	RN
[84] [b]	WH	CL-VS	PT (SS)	n/c	n/a			2 days	LP	RN
[85] [a]	WHT	n/s				n/c	n/a	192 h–24 h	AHH	n/c
[86]	WHT	n/s				RR (MS)	n/c	1 day	MILP	RN
[88]	WHT	CL-VS				RR	n/c	1 day	MILP	n/c
[89] [a]	WHT	CL-FS				n/c	n/a	1 year	AHH	n/c
[90]	WH	CL-FS				n/c	n/a	1 day	AHH	n/a
[91]	WH	CL-FS				n/c	n/a	1 year	AHH	n/c
[92] [d]	PSHP	CL-FS	PT (SS)	n/c	n/a			1 day	MILP	RN
[98] [f]	HT	CL-VS				n/c	n/a	1 week	GA+TS	RN
[99]	PSHP	CL-VS	PT (SS)	PT (SS)	FP			1 day	MILP	n/c
[112] [a]	WHT	CL-FS				RR (SS)	RW	1 day UC+1 h ED	PRM+LP	n/c
[115]	PSHP	OL-FS	PT (SS)	PT (MS)	FP			1 year	SDP+MILP	RN

[a] The paper is more focused on the analysis (role of the PSHP, optimal sizing, etc.) rather than on modeling.
[b] A continuous range of power is considered in pumping mode even though the concept of variable speed operation is not mentioned in the paper.
[c] Real time use of committed reserves is used to calculate the income but it is not used in the balance equation.
[d] Wind power is considered as a exogenous stochastic variable.
[e] It is not clear whether or not a dump variable is used to allow for wind curtailments.
[f] Thermal generating units are committed by means of a DP-LP based algorithm.

Consistently, special attention is being given to accurately modeling the uncertainty in energy and ancillary services prices, as well as in wind and/or solar power production, for the power generation scheduling of PSHPs in both liberalized and centralized market contexts.

By means of a proper planning and operation scheduling, PHES can help to reduce the power supply costs and/or electricity prices, to integrate RENs in the power system and to reduce greenhouse gases emissions from fossil fuel-fired power plants.

Finally, the papers above discussed within which authors use or propose a power generation and consumption scheduling model for a PSHP or a set of PSHPs (considered individually or as a part of a bigger power generation system), for whatever purpose, are classified in Table 1, according to a set of criteria which are related to the above described trends. The column "Generation technologies" refers to the generation technologies considered in the model, whose generated powers are considered as endogenous variables. In those cases where a dump or slack variable is used to allow for possible wind power curtailments or deviations with respect to the forecast wind power, wind power injected into the power system or offered in the corresponding market has been considered as an endogenous variable. The column "Type of PSHP" refers to whether or not the PSHP receives natural water inflows in the upper reservoir and to the use of fixed or

Table 2
Summary of challenges and papers where they have been preliminarily addressed.

Challenge	Papers where it has been preliminarily addressed
3.1 Considering the uncertainty of the real time use of regulation reserves [a]	[51], [64], [69], [70], [71], [112]
3.2 Revising the methods currently used to determine the long-term guidelines	[72], [115]
3.3 Defining the optimal strategy for PSHPs with variable speed and hydraulic short-circuit operation	[70], [88], [98], [99], [101]

[a] In [29,38,99,115], a fixed percentage of the committed reserve, not explicitly related with the probability or the average value, is assumed to be used in real time.

variable speed operation. The columns "Spot market" and "Reserve markets" refer to whether or not the behavior of the other competitors in the corresponding markets is considered in the paper, as well as to how market prices are modeled. It is important to note that the term "price-taker" does not necessarily correspond to "price-acceptant" (i.e. the price-taker may consider the bid price as an endogenous variable), and that the column "Reserve markets" refers to any type of reserve market. The column "Regulation reserve" refers to whether or not and how the regulation reserve is considered in the papers where a centralized market approach is used. The columns "Real time of committed reserves" refers to whether or not and how the use of committed reserves is considered in the paper, both in liberalized and centralized market contexts. The columns "Time horizon" and "Operation scheduling algorithm" refer respectively to the duration of the time horizon and the type of algorithm of the scheduling model used or proposed in the paper. It should be noted that the time horizon of the scheduling model does not necessarily coincide with neither the decision horizon of the model (some authors use a look-ahead period) nor the time horizon of the study presented in the paper. The column "Risk aversion" refers to whether or not and how risk aversion is considered in the paper. The meaning of each symbol used in the table can be found in the Appendix.

3. Challenges

The main challenges from the perspective of PSHP operators, which have been observed by the authors when carrying out the literature review to prepare the paper, can be summarized as (1) to improve the techniques to model the possible usage by the TSO of the offered regulation reserves, (2) to revise the methods traditionally used to determine the so-called long-term guidelines, such as the target stored volume at the end of the day or week, as well as the marginal water value functions, by considering the participation of the PSHP in the reserve markets, and (3) to explore the new bidding and scheduling strategies when the PSHP increases its flexibility for providing regulation control thanks to variable speed, or hydraulic-short circuit configurations. At the end of Section 3, the above-mentioned challenges, along with the papers where they have been preliminarily addressed are summarized in Table 2. Hereinafter, an explanation of the said challenges of them is provided.

3.1. Considering the uncertainty of the real time use of regulation reserves

Considering the uncertainty in the power delivery requested in real time for ancillary services purposes is still a pending task that, according to authors' experience, could contribute significantly to increasing the participation of PSHPs in the said services, and thus the penetration of intermittent energies into the electric power system, and to improving the PSHPs economic feasibility.

There exist plenty of spot market prices forecasting models in the technical literature, with a noteworthy prediction accuracy [45,105,106], as well as quite a few balancing market prices forecasting models [107,108]. Nevertheless, to authors' knowledge, there is no documented attempt to forecast the real time use of committed power reserves, even though as it was above-mentioned, there are many electricity markets all over the world where balancing services are remunerated for two different concepts, availability and delivery.

In most of the papers revised for writing this paper, authors either do not consider the amount of committed power reserve requested in real time for balancing purposes [59], or use a single hourly value [51,38]. For open-loop PSHPs, associated to large reservoirs, it seems reasonable to think that the effectively used reserves do not have a significant impact on the available head or stored energy [37]. Nevertheless, they may prevent the PSHP from fulfilling the end of week target stored energy or affect the economic value of water stored in the reservoir at the end of the week [109]. For closed-loop PSHPs, usually associated to a small reservoir, the uncertainty in the effectively used power reserve may not only affect the fulfilment of the end of day or week target stored energy, but also the fulfilment of the operation schedule committed in the spot market, with the corresponding economic penalty. Consequently, an accurate forecast of the real time use of committed power reserves can be of considerable importance for close-loop PSHPs.

As it was above-mentioned, the amount of power requested in real time for balancing purposes depend on real time power unbalances [28], and these can be caused by very diverse reasons. It seems logical that in power systems with higher shares of non-dispatchable energy, the need of balancing services is higher, and vice versa, in power systems with predominance of dispatchable energy, such as hydropower (Norway, Canada), the need of balancing services is expected to be lower, in relative terms [44]. In mainland Spain, for example, where wind installed power is 22% of total installed power, the percentage of committed power reserves requested in real time for balancing purposes is significant (Fig. 1): in 2010 the mean percentages of upward and downward regulation reserve requested in real time for balancing purposes 28.6% and 30.0%, respectively.

The influence of the wind power forecast on the spot market prices has been already analyzed in a few papers [110,111]. As explained in [112], the uncertainty level in wind power forecast is influenced, among other things, by the wind penetration level. Therefore, it seems reasonable to think that the wind power forecast might have a significant influence on the real time use of committed reserve, and as a consequence, on the prices of committed and delivered power reserve. Even though, there have been studies where the wind power forecast is used to determine the power reserve requirements in power systems with significant wind power penetration [113,114], as far as we know, the analysis of the influence of wind power forecast on the power reserve prices and real time use of committed reserves has not been addressed yet in a market context where the provision of power reserves depends on the utilities' decisions.

3.2. Revising the methods currently used to determine the long-term guidelines (target stored volumes and marginal water value functions)

The determination of the long-term guidelines typically used as a link between the short- and long-term scheduling for open-loop PSHPs (storage targets and marginal water values), as well as of

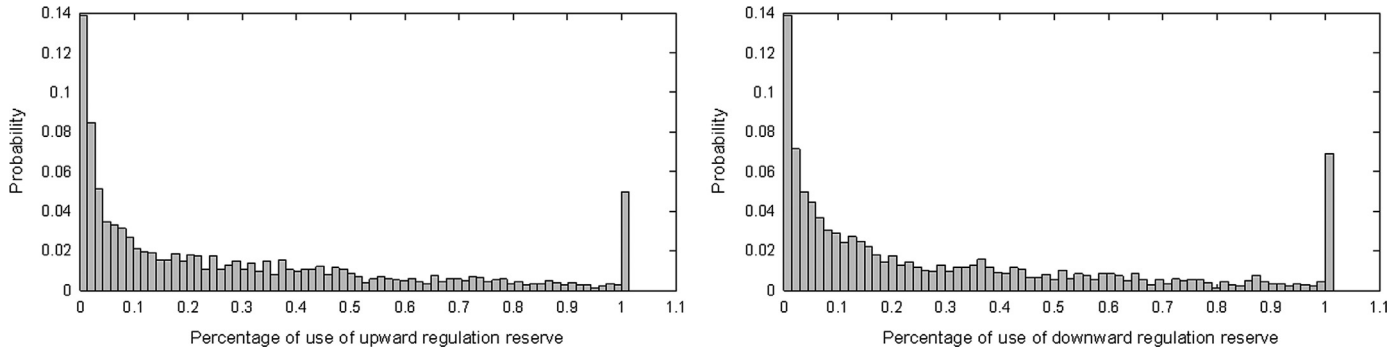


Fig. 1. Percentage of use of upward and downward regulation reserve in the mainland Spanish power system (2010).

the end of day or week storage targets traditionally used for closed-loop PSHPs, should be revisited.

The traditional "empty at 0:00" target used for daily-cycle PSHPs prevents the PSHP from providing upward spinning reserve during first hours of each day, during which there is usually a big amount of base power plants in operation, with little and/or inefficient power regulation capabilities, like in the French, German or Finnish power systems, and during which the consequences of abrupt power variations such as those caused by unexpected load rejections, atmospheric phenomena, etc., can be more severe due to the power system's lower inertia.

Authors guess that either the use of different end of day or week targets "updated" to the new reserve-driven operation strategy, or the replacement of such "targeted" operation strategy by a flexible one with a suitable look-ahead period, similar to the one used in [72], would yield better results in terms of both the PSHP profitability and participation in the ancillary services.

In order to update the above-mentioned targets, authors would suggest, as a first approach, to run a multi-market scheduling or bidding model, with a longer time horizon (e.g. 1 week for daily-cycle PSHPs; several weeks for weekly-cycle PSHPs), using several characteristic historical scenarios beginning on different days of the week. Stored energy in the upper reservoir would be constrained only at the beginning and the end of the time horizon. From the results of this simple optimization exercise, it would be possible to gain insight on the end of day and week targets which allow the PSHP making a bigger profit from the considered energy and ancillary services markets, under the assumption of a perfect knowledge of the next day or week energy and ancillary services prices. Further conclusions could be drawn by using different energy and ancillary prices series within each day of the above-described approach, and a stochastic version (or rather deterministic equivalent) of the above-mentioned multi-market scheduling model.

For open-loop PSHPs, authors believe that the methodology presented in [115] may be of help for determining the end of week storage targets or marginal water value functions, considering the participation of the PSHP both in energy and ancillary services markets. In that paper, authors considered the participation of a large open-loop PSHP in the secondary load-frequency control market within a long term scheduling model. For this purpose, authors use a stochastic dynamic programming (SDP) based model, with a 1-year time horizon, divided into weekly stages. Water level or storage in the reservoir at the beginning of each week is used as state variable. Weekly water inflows are considered in a deterministic way. Several scenarios of hourly pool prices and weekly secondary control prices are considered within each week. The work is carried out within the framework of the Swiss electric power system, where weekly "capacity blocks" (in MW) are submitted to the TSO (Swissgrid) for secondary control purposes. For each water level and secondary control price, the expected weekly profit is calculated by means of a MILP based weekly scheduling model. The expected future profit is then calculated by solving a simple recursive equation for each feasible state; water values are obtained from the derivative of the expected future profit values.

Even though the results obtained in the paper showed a small sensitive of the long-term scheduling to the consideration of the secondary control offers (1–2% increase in the expected profit when secondary control is considered), similar analyses should be carried out with other PSHP and electricity markets. The structure of the secondary control market in the Swiss power system might introduce certain degree of "rigidity" in the PSHP operation, which may not appear in other secondary reserve markets with hourly auctions.

For open-loop PSHPs integrated in large hydropower systems, a SDDP based approach should be used to determine the marginal water value functions in order to avoid the so-called curse of dimensionality. The model presented in [80] may be used for this purpose. As it was above-mentioned, in that paper authors considered the bidding decisions in the day-ahead and intraday markets, with a hourly time resolution, in a long-term hydro scheduling model. Instead of, or in addition to the intraday markets, the reserve or balancing markets could be considered in the model. Should the computational burden of the problem become intractable, the approximation proposed in [82] to model the start-up costs of generating units by means of continuous variables might be useful.

3.3. Defining the optimal strategy for PSHPs with variable speed and hydraulic short-circuit operation

Taking into account the ongoing PSHPs projects all over the world, further studies on the operation strategy of more flexible PSHPs, such as those using variable speed equipment or operating in hydraulic short-circuit mode, become mandatory.

As it was above-mentioned, to authors' knowledge, there are only three PSHPs currently operating in short-circuit mode: Kops II, Malta and Geesthacht. Kops II is composed of three ternary units, each comprising one Pelton turbine connected through a lock-up clutch to a pump, both coupled to a synchronous machine. Thanks to the clutch, each unit can operate simultaneously in pumping and generating mode; the pumps operating at a fixed consumption power level. In this way, the power plant can be "seen" by the grid as a controllable load, with a power regulation range equal to that of the turbines in operation.

From the modeling point of view, it is important to take into account that these PSHPs have an additional operating mode in comparison with conventional PSHPs, the hydraulic short-circuit mode, hereinafter referred to as SC-mode.

Generating, pumping and SC-mode power-discharge curves are shown in Figs. 2 and 3 for a PSHP with a single ternary unit, assuming

that head variation is much lower than rated gross head. Generating mode refers to the case when the turbine is on and the pump is off, and vice versa, pumping mode refers to the case when the pump is on and the turbine is off. SC-mode curve can be obtained by means of a simple translation of generating mode curve (Fig. 2, 1st quadrant); i.e. subtracting to every point in the generating mode curve maximum pump water flow and power consumption. The available regulation-up and down reserves (Reg-up R. and Reg-dw R., respectively) in SC-mode are shown in Fig. 3. Regulation-up energy is delivered by increasing the water flow discharged through the turbine (pumped water flow does not change) while regulation-down energy is obtained by reducing water flow discharged through the turbine.

When the PSHP comprises n ternary units (with $n > 1$), it is possible to operate simultaneously np pumps and nt turbines, with $0 \le np \le n$ and $0 \le nt \le n$, with a total power regulation range equal to that of the turbines in operation.

Theoretically, PSHPs composed of n binary units, $n \ge 2$, (reversible pump-turbine coupled to a synchronous machine) can also operate in SC-mode. Nevertheless, the power regulation range in SC-mode is considerably lower than with ternary units, since $np + nt \le n$.

Variable speed operation is a well-known solution for enlarging flexibility of a PSHP. The first variable speed PSHPs were commissioned in Japan at the beginning of the nineties of the past century. Most variable speed pump-turbine units are equipped with doubly-fed induction machines, due to the fact that they have a better overall performance than the synchronous machines, and that they need a lower converter rating [116]. Unlike conventional PSHPs, variable speed PSHPs are capable to provide power regulation in both operating modes. In a conventional PSHP, the operating conditions of the unit in pumping mode are represented by the solid lines in Fig. 4 (n_{des}), where solid lines and broken lines represent the performance curve at the design rotation speed (n_{des}) and at the minimum and maximum rotation speed (n_{min} and n_{max}) of the unit respectively. The dotted area represents the operating range of the variable speed unit.

As it can be seen in the figure, at a fixed head a fixed-speed unit is only able to consume a fixed power g^p. The possibility of varying the rotation speed does not only increase the operating range of the unit (dotted area) but also allows the unit to vary its power consumption, even at a fixed head (see Fig. 4, from $\underline{g^p}$ to $\overline{g^p}$), and hence to provide power regulation in pumping mode, resulting in the power-discharge curves reported in Fig. 5.

Additionally, variable speed operation allows increasing the efficiency and the head and flow operating range in generating mode [97]. Nevertheless, since PSHPs are usually located in large

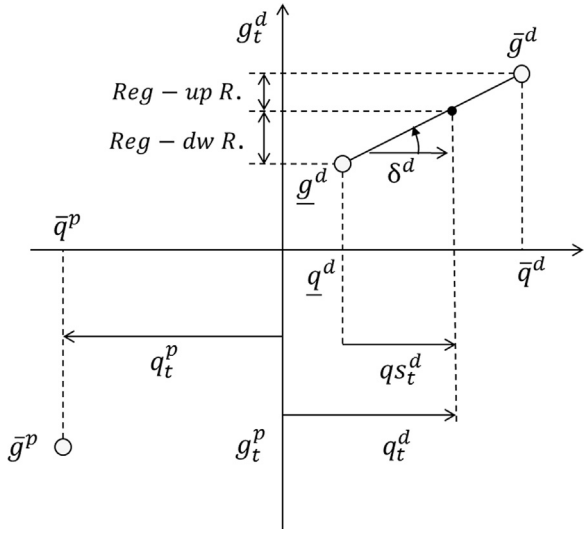


Fig. 2. Power-discharge curve of a ternary unit in generating and pumping modes.

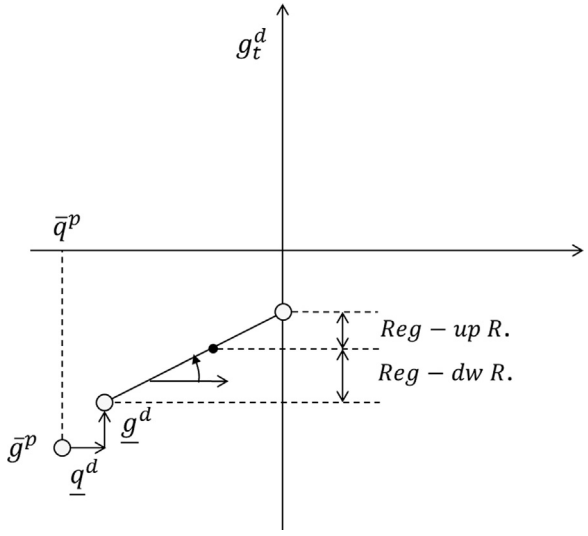


Fig. 3. Power-discharge curve of a ternary unit in short-circuit mode.

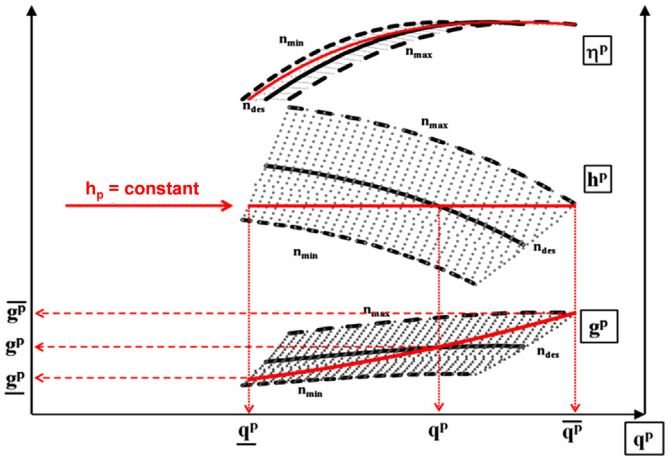


Fig. 4. Performance curve of variable speed pump-turbine unit in pumping mode: efficiency, head and pumping power.

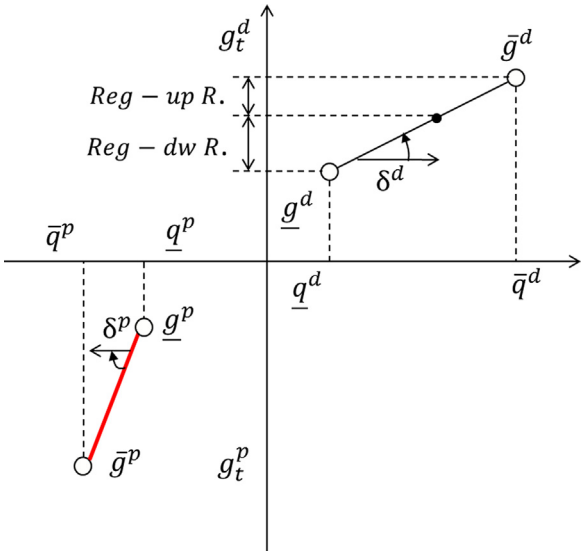


Fig. 5. Power-discharge curve of a variable speed pump-turbine unit in generating and pumping modes.

head sites, the increase in operating range and efficiency in generating mode is not very significant.

From the modeling point of view, variable speed PSHPs also have some particularities which are worthy to be mentioned. A typical power-discharge curve of a variable speed pump-turbine is shown in Fig. 5, with the mentioned power regulation range in pumping mode; Reg-up R. when pumping is the difference between minimum and actual consumed power while Reg-dw R. is the difference between actual and maximum consumed power.

Variable PSHPs with n units ($n \geq 2$) perform in an analogous way. The bigger the number of units, the wider the available power regulation range both in generating and pumping modes, provided that each unit is coupled to the grid through a proper frequency converter. In the case that the PSHP is equipped with m converters ($m < n$), the model formulation shall take into account that, as a maximum, m units will be able to regulate power simultaneously in pumping mode.

4. Conclusions

This paper has highlighted the crucial role that pumped-storage can play in a framework of high level penetration of wind power since it helps to increase the amount of wind power eventually integrated in the power system, both in deregulated and centralized market contexts. Additionally, this paper shows that ancillary services markets, particularly those related to balancing supply and demand such as the regulating reserve markets, emerge as a valuable source of revenue for PSHPs. In this context of new balancing or reserve-driven operation strategies, traditional operation rules of pumped-storage units could be no longer valid (or at least not always).

Current trends in the operation of PSHPs have been properly described and discussed, and three main challenges to be dealt with in the next future have been identified. Additionally, some approaches to address the identified challenges have been suggested. The first identified challenge from the perspective of PSHPs operators is to model in a more accurate way the possible usage by the SO of the offered regulation reserves (upward and downward), as they have a direct influence on the physical operation of the plant and on the stored water in the reservoirs. Additionally, given that hedging against the exposure to risk requires an accurate modeling of energy and ancillary services prices, it has been identified that there is a lack of studies where the influence of the PSHP energy and reserve bids on the resulting market clearing prices is properly taken into account.

The second identified challenge is to revise the methods currently used to determine the so-called long-term operation guidelines, which are usually given in the form of target stored volumes at the end of the day/week and marginal water value functions. These long-term signals have been traditionally used in the operation scheduling of PSHPs in the past, and in case of increasing the participation in regulation reserve markets, it is clear that more sophisticated criteria need to be elaborated to fully take advantage of the joint operation both in the energy and in the reserve markets.

Finally, the third identified challenge is related to the lack of strategic bidding models and/or market-oriented scheduling tools that consider the capability of providing regulation reserve while pumping in case of variable speed or hydraulic-short circuit configurations. All the models found in the literature do not consider the flexibility that these two configurations can provide, and therefore new mathematical developments are needed in the future.

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Appendix. Acronyms used throughout the paper (in alphabetical order):

ADDP	approximate dual dynamic programming
AHH	ad hoc heuristic algorithm based on a set of operation rules
BD	benders decomposition
CL	closed-loop PSHP (with no natural inflows in the upper reservoir)
CVaR	*Conditional Value at Risk* (in Table 1 it means that CVaR is used to hedge against the exposure to risk)
DP	dynamic programming
DR	the *downside risk* is used to hedge against the exposure to risk
ED	economic dispatch
EEX	European energy exchange spot market
FP	fixed percentage or probability of use
FS	fixed speed operation
GA	genetic algorithm
HS	hydropower system
HT	hydrothermal system
i/c	implicitly considered
LP	linear programming
LR	Lagrangian relaxation
MILP	mixed integer linear programming
MINLP	mixed integer nonlinear programming
MS	multiple scenarios
n/a	not applicable
n/c	not considered in the paper
NEPOOL	New England power pool
NLP	nonlinear programming
n/s	not specified in the paper
NYISO	New York independence system operator
OL	open-loop PSHP (with natural inflows in the upper reservoir)
PHES	pumped-hydro energy storage
PM	price-maker (i.e. the behavior of other competitors is considered)
PRM	priority ranking method
PT	price-taker (i.e. the behavior of other competitors is not considered)
PSHP	pumped-storage hydropower plant
RDC	residual demand curves
Reg-dw R.	regulation down reserves
Reg-up R.	regulation up reserves
RES	renewable energy sources
RH	rolling horizon
RN	risk-neutral approach to hedge against the exposure to risk (i.e. scenario-based approach)
RR	reserve requirement imposed on a single or in multiple scenarios

RW	imbalances between actual and forecast wind power is simulated as a random walk
SC-mode	hydraulic short-circuit operating mode
SDP	stochastic dynamic programming
SDDP	stochastic dual dynamic programming
SS	single scenario
TS	tabu search
TSO	transmission system operator
UC	unit commitment
VaR	*Value at Risk*
VFE	the variance of the forecast errors is used to hedge against the exposure to risk
VS	variable speed operation
WH	wind-hydro system
WHT	wind-hydro-thermal power system

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